Attention Is All You Need

7 claims not supported · 2 references need attention.

Claims: 7 not supported · 32 partly supported · 12 not enough evidence · 7 supported · 21 need review

References: 1 details differ · 1 could not check · 38 verified

Audited on 13 November 2026 · Shareable link: `http://127.0.0.1:8797/r/demo`

Claims · 58

Needs your attention · 24

Not supported needs review

Most competitive neural sequence transduction models have an encoder-decoder structure [5,2,35].

"The proposed neural network architecture, which we will refer to as an RNN Encoder-Decoder, consists of two recurrent neural networks (RNN) that act as an encoder and a decoder pair."

The judge and the skeptic disagree; the more cautious verdict is shown.

c11 → ref b4 · full text · Cho et al. 2014, Learning phrase representations using rnn encoder-decoder for statistical machine translation · source (https://aclanthology.org/D14-1179.pdf)

Not supported needs review

We used beam search with a beam size of 4 and length penalty α = 0.6 [38].

"Typically we use beamsize=3.0, unless otherwise noted."

The judge and the skeptic disagree; the more cautious verdict is shown.

c29 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Not supported needs review

Furthermore, RNN sequence-to-sequence models have not been able to attain state-of-the-art results in small-data regimes [37].

"We trained a sequence-to-sequence model with attention on the small human-annotated parsing dataset and were able to achieve an F1 score of 88.3 on section 23 of the WSJ without the use of an ensemble and 90.5 with an ensemble, which matches the performance of the BerkeleyParser (90.4) when trained on the same data."

The judge and the skeptic disagree; the more cautious verdict is shown.

c32 → ref b36 · full text · Vinyals et al. 2015, Grammar as a foreign language · source (https://arxiv.org/html/1412.7449)

Not supported needs review

We also trained it in a semi-supervised setting, using the larger high-confidence and BerkleyParser corpora from with approximately 17M sentences [37].

"We call the set of \sim11 million sentences selected in this way, together with the \sim90K golden sentences described above, the high-confidence corpus. In earlier experiments, we only used one parser, our reimplementation of BerkeleyParser, to create a corpus of parsed sentences. In that case we just parsed \sim7 million senteces from news appearing on the web and combined these parsed sentences with the \sim90K golden corpus described above."

The judge and the skeptic disagree; the more cautious verdict is shown.

c34 → ref b36 · full text · Vinyals et al. 2015, Grammar as a foreign language · source (https://arxiv.org/html/1412.7449)

Not supported

In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17,18] and [9].

"A number of neural architectures have been proposed for modelling translation, such as encoder-decoder networks (Kalchbrenner & Blunsom, 2013; Sutskever et al., 2014; Cho et al., 2014; Kaiser & Bengio, 2016), networks with attentional pooling (Bahdanau et al., 2014) and two-dimensional networks (Kalchbrenner et al., 2016a)."

The cited source is the ByteNet paper on neural machine translation.

c10 → ref b17 · full text · Kalchbrenner et al. 2017, Neural machine translation in linear time · source (https://arxiv.org/html/1610.10099)

Not supported

In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17,18] and [9].

"We achieve a new state of the art on several public translation benchmark data sets."

The cited source is a convolutional seq2seq paper (Gehring et al.), not the Transformer paper.

c10 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Not supported

In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence length n is smaller than the representation dimensionality d, which is most often the case with sentence representations used by state-of-the-art models in machine translations, such as word-piece [38] and byte-pair [31] representations.

"Character n-grams allow for different trade-offs between sequence length (# tokens) and vocabulary size (# types), depending on the choice of n."

The source is the byte-pair encoding paper.

c21 → ref b30 · full text · Sennrich et al. 2015, Neural machine translation of rare words with subword units · source (https://arxiv.org/pdf/1508.07909)

Partly supported needs review

Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38,24,15].

"Since then, many novel techniques have been proposed to further improve NMT"

The judge and the skeptic disagree; the more cautious verdict is shown.

c2 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Partly supported needs review

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions.

"The use of operations that run in parallel along the sequence length can also be beneficial for reducing computation time."

The judge and the skeptic disagree; the more cautious verdict is shown.

c6 → ref b17 · full text · Kalchbrenner et al. 2017, Neural machine translation in linear time · source (https://arxiv.org/html/1610.10099)

Partly supported needs review

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions.

“Compared to recurrent models, computations over all elements can be fully parallelized during training to better exploit the GPU hardware”

The judge and the skeptic disagree; the more cautious verdict is shown.

c6 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Partly supported needs review

End-to-end memory networks are based on a recurrent attention mechanism instead of sequencealigned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [34].

“a neural network with an explicit memory and a recurrent attention mechanism for reading the memory can be successfully trained via backpropagation on diverse tasks from question answering to language modeling”

The judge and the skeptic disagree; the more cautious verdict is shown.

c9 → ref b33 · full text · Sukhbaatar et al. 2015, End-to-end memory networks · source (https://arxiv.org/html/1503.08895)

Partly supported needs review

Most competitive neural sequence transduction models have an encoder-decoder structure [5,2,35].

“Most of the proposed neural machine translation models belong to a family of encoder–decoders”

The judge and the skeptic disagree; the more cautious verdict is shown.

c11 → ref b1 · full text · Bahdanau et al. 2014, Neural machine translation by jointly learning to align and translate · source (https://arxiv.org/html/1409.0473)

Partly supported needs review

We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization [1].

“We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously.”

The judge and the skeptic disagree; the more cautious verdict is shown.

c13 → ref b10 · abstract only · He et al. 2016, Deep residual learning for image recognition · source (https://doi.org/10.1109/cvpr.2016.90)

Partly supported needs review

This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [38,2,9].

“Our model (see Figure 1) follows the common sequence-to-sequence learning framework [41] with attention [2].”

The judge and the skeptic disagree; the more cautious verdict is shown.

c16 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Partly supported needs review

This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [38,2,9].

"The dominant approach to date encodes the input sequence with a series of bi-directional recurrent neural networks (RNN) and generates a variable length output with another set of decoder RNNs, both of which interface via a soft-attention mechanism"

The judge and the skeptic disagree; the more cautious verdict is shown.

c16 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Partly supported needs review

In our model, we share the same weight matrix between the two embedding layers and the pre-softmax linear transformation, similar to [30].

"For translation models, we also present a three-way weight tying method."

The judge and the skeptic disagree; the more cautious verdict is shown.

c17 → ref b29 · full text · 2016, Using the output embedding to improve language models · source (https://arxiv.org/html/1608.05859)

Partly supported needs review

We employ three types of regularization during training: Residual Dropout We apply dropout [33] to the output of each sub-layer, before it is added to the sub-layer input and normalized.

"This significantly reduces overfitting and gives major improvements over other regularization methods."

The judge and the skeptic disagree; the more cautious verdict is shown.

c27 → ref b32 · abstract only · Srivastava et al. 2014, Dropout: a simple way to prevent neural networks from overfitting · source (https://openalex.org/W2095705004)

Not enough evidence needs review reference differs

Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4,27,28,22].

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

Cited reference: details differ from the found record (title).

c8 → ref b26 · full text · Parikh et al. 2016, A decomposable attention model · source (https://www.aclweb.org/anthology/D16-1244.pdf)

Not enough evidence needs review

There are many choices of positional encodings, learned and fixed [9].

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c18 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Not enough evidence needs review

We also experimented with using learned positional embeddings [9] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)).

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c19 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Not enough evidence needs review

In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence length n is smaller than the representation dimensionality d, which is most often the case with sentence representations used by state-of-the-art models in machine translations, such as word-piece [38] and byte-pair [31] representations.

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c21 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Not enough evidence needs review

For English-French, we used the significantly larger WMT 2014 English-French dataset consisting of 36M sentences and split tokens into a 32000 word-piece vocabulary [38].

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c25 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Not enough evidence needs review

Label Smoothing During training, we employed label smoothing of value ϵ ls = 0.1 [36].

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c28 → ref b35 · abstract only · Szegedy et al. 2015, Rethinking the inception architecture for computer vision · source (https://doi.org/10.1109/cvpr.2016.308)

Not enough evidence needs review

In row (E) we replace our sinusoidal positional encoding with learned positional embeddings [9], and observe nearly identical results to the base model.

No verbatim quote from the source backs a verdict.

The judge and the skeptic disagree; the more cautious verdict is shown.

c31 → ref b8 · full text · Gehring et al. 2017, Convolutional sequence to sequence learning · source (https://arxiv.org/pdf/1705.03122)

Partly supported · 22

Partly supported

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35,2,5].

“We briefly review Hochreiter's (1991) analysis of this problem, then address it by introducing a novel, efficient, gradient based method called long short-term memory (LSTM).”

The source is the original LSTM paper, so it backs the citation for LSTM as a recurrent architecture.

c1 → ref b12 · abstract only · Hochreiter et al. 1997, Long short-term memory · source (https://doi.org/10.1162/neco.1997.9.8.1735)

Partly supported

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35,2,5].

“almost none of these successes were achieved with a vanilla recurrent neural network. Rather, it was a recurrent neural network with sophisticated recurrent hidden units”

The source (Chung et al. 2014) says recent successes in tasks like machine translation came from RNNs with gated units such as LSTM.

c1 → ref b6 · full text · Chung et al. 2014, Empirical evaluation of gated recurrent neural networks on sequence modeling · source (https://arxiv.org/html/1412.3555)

Partly supported

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35,2,5].

“Our main result is that on an English to French translation task from the WMT’14 dataset, the translations produced by the LSTM achieve a BLEU score of 34.8 on the entire test set”

The source (Sutskever et al., seq2seq) shows LSTMs achieving strong results on machine translation and describes an RNN language model.

c1 → ref b34 · full text · Sutskever et al. 2014, Sequence to sequence learning with neural networks · source (https://arxiv.org/html/1409.3215)

Partly supported

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35,2,5].

“Sutskever et al. (2014) reported that the neural machine translation based on RNNs with long short-term memory (LSTM) units achieves close to the state-of-the-art performance of the conventional phrase-based machine translation system”

This source (Bahdanau et al., neural machine translation) shows LSTM/gated-unit RNNs applied to machine translation with promising results.

c1 → ref b1 · full text · Bahdanau et al. 2014, Neural machine translation by jointly learning to align and translate · source (https://arxiv.org/html/1409.0473)

◐ Partly supported

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35,2,5].

“In the field of statistical machine translation (SMT), deep neural networks have begun to show promising results.”

The paper uses RNN encoder-decoders for machine translation and reports improvements.

c1 → ref b4 · full text · Cho et al. 2014, Learning phrase representations using rnn encoder-decoder for statistical machine translation · source (https://aclanthology.org/D14-1179.pdf)

◐ Partly supported

Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38,24,15].

“we hope to inspire research on large scale Language Modeling, a problem we consider crucial towards language understanding”

The source is a study that pushes the limits of recurrent language models (LSTMs, character CNNs), which fits the "recurrent language models" part of the claim.

c2 → ref b14 · full text · Jozefowicz et al. 2016, Exploring the limits of language modeling · source (https://arxiv.org/html/1602.02410)

◐ Partly supported

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2,19].

“allowing a model to automatically (soft-)search for parts of a source sentence that are relevant to predicting a target word”

The source (Bahdanau et al.) introduces an attention mechanism for neural machine translation, which supports attention being used in sequence transduction.

c4 → ref b1 · full text · Bahdanau et al. 2014, Neural machine translation by jointly learning to align and translate · source (https://arxiv.org/html/1409.0473)

◐ Partly supported

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2,19].

“Attention networks are now a standard part of the deep learning toolkit, contributing to impressive results in neural machine translation”

The source says attention networks are a standard part of deep learning and have helped in many tasks, including machine translation.

c4 → ref b18 · full text · Kim et al. 2017, Structured attention networks · source (https://arxiv.org/html/1702.00887)

◐ Partly supported

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions.

“When the issue of recurrent output dependencies is addressed, as we do in the Extended Neural GPU, an active memory model can indeed match or exceed attention models”

The source introduces the Extended Neural GPU as a convolutional model that processes the whole input in parallel.

c6 → ref b15 · full text · Kaiser et al. 2016, Can active memory replace attention? · source (https://arxiv.org/html/1610.08613)

◐ Partly supported

This makes it more difficult to learn dependencies between distant positions [12].

“error signals "flowing backwards in time" tend to either (1) blow up or (2) vanish”

The source says backpropagated error signals in recurrent networks blow up or vanish, and that learning is hard when time lags between inputs and teacher signals are long.

c7 → ref b11 · abstract only · Hochreiter et al. 2001, Gradient flow in recurrent nets: the difficulty of learning long-term dependencies · source (https://openalex.org/W1525783482)

◐ Partly supported

Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4,27,28,22].

“We introduce a neural network model with a novel intra-attention that attends over the input and continuously generated output separately”

The source (Paulus et al.) uses intra-attention (a form of self-attention) successfully for abstractive summarization, which backs only that one task.

c8 → ref b27 · full text · Paulus et al. 2017, A deep reinforced model for abstractive summarization · source (https://arxiv.org/html/1705.04304)

◐ Partly supported

Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4,27,28,22].

“We evaluate our model on 3 different tasks: author profiling, sentiment classification and textual entailment.”

The source is the Lin et al. self-attentive sentence embedding paper, which uses self-attention for textual entailment and learns task-independent sentence representations.

c8 → ref b21 · full text · Lin et al. 2017, A structured self-attentive sentence embedding · source (https://arxiv.org/html/1703.03130)

◐ Partly supported

Most competitive neural sequence transduction models have an encoder-decoder structure [5,2,35].

“32M for the “encoder” LSTM and 32M for the “decoder” LSTM”

The source uses an encoder-decoder LSTM architecture that achieves strong translation results.

c11 → ref b34 · full text · Sutskever et al. 2014, Sequence to sequence learning with neural networks · source (https://arxiv.org/html/1409.3215)

◐ Partly supported

The two most commonly used attention functions are additive attention [2], and dot-product (multiplicative) attention.

“Intuitively, this implements a mechanism of attention in the decoder.”

The source (Bahdanau et al.) introduces the attention mechanism that is later called additive attention, but the excerpts never use that term.

c14 → ref b1 · full text · Bahdanau et al. 2014, Neural machine translation by jointly learning to align and translate · source (https://arxiv.org/html/1409.0473)

◐ Partly supported

While for small values of d k the two mechanisms perform similarly, additive attention outperforms dot product attention without scaling for larger values of d k [3].

“We found that the parameterized additive attention mechanism slightly but consistently outperformed the multiplicative one”

The source finds additive attention slightly but consistently better than multiplicative attention.

c15 → ref b2 · full text · Britz et al. 2017, Massive exploration of neural machine translation architectures · source (https://www.aclweb.org/anthology/D17-1151.pdf)

◐ Partly supported

The shorter these paths between any combination of positions in the input and output sequences, the easier it is to learn long-range dependencies [12].

“error signals "flowing backwards in time" tend to either (1) blow up or (2) vanish”

The source is about vanishing and exploding gradients in recurrent networks, which is the basis for why long-range dependencies are hard to learn.

c20 → ref b11 · abstract only · Hochreiter et al. 2001, Gradient flow in recurrent nets: the difficulty of learning long-term dependencies · source (https://openalex.org/W1525783482)

◐ Partly supported

Doing so requires a stack of O(n/k) convolutional layers in the case of contiguous kernels, or O(log k (n)) in the case of dilated convolutions [18], increasing the length of the longest paths between any two positions in the network.

“The dilated structure of the convolutions connects any two source or target tokens in the sequences by way of a small number of network layers”

ByteNet supports the claim that dilated convolutions reach long dependencies in few layers, with a constant c ≈ log d.

c22 → ref b17 · full text · Kalchbrenner et al. 2017, Neural machine translation in linear time · source (https://arxiv.org/html/1610.10099)

◐ Partly supported

We used the Adam optimizer [20] with β 1 = 0.9, β 2 = 0.98 and ϵ = 10 -9 .

“Good default settings for the tested machine learning problems are α = 0.001, β 1 = 0.9, β 2 = 0.999 and = 10 -8 .”

The source is the Adam paper, so it supports citing Adam as the optimizer and β1 = 0.9.

c26 → ref b19 · full text · Kingma et al. 2015, Adam: A method for stochastic optimization · source (https://arxiv.org/pdf/1412.6980)

Partly supported

We set the maximum output length during inference to input length + 50, but terminate early when possible [38].

“where we simply stop decoding after a predetermined maximum output length of twice the source length”

The source (GNMT) supports early termination through pruning once a good hypothesis is found.

c30 → ref b37 · full text · Wu et al. 2016, Google's neural machine translation system: Bridging the gap between human and machine translation · source (https://arxiv.org/html/1609.08144)

Partly supported

We trained a 4-layer transformer with d model = 1024 on the Wall Street Journal (WSJ) portion of the Penn Treebank [25], about 40K training sentences.

“a corpus of over 4 million words of running text annotated with part-of- speech (POS) tags, with over 3 million words of that material assigned skeletal grammatical structure”

The source describes the Penn Treebank corpus, so the citation is appropriate for the dataset.

c33 → ref b24 · abstract only · Marcus et al. 1993, Building a large annotated corpus of english: The penn treebank · source (https://doi.org/10.21236/ada273556)

Partly supported

Our results in Table 4 show that despite the lack of task-specific tuning our model performs surprisingly well, yielding better results than all previously reported models with the exception of the Recurrent Neural Network Grammar [8].

“Our generative model obtains (i) the best-known parsing results using a single supervised generative model”

The source (the RNNG paper) reports strong parsing results, but the excerpts do not show the citing paper's Table 4 or its comparisons.

c35 → ref b7 · full text · Dyer et al. 2016, Recurrent neural network grammars · source (https://www.aclweb.org/anthology/N16-1024.pdf)

Partly supported

In contrast to RNN sequence-to-sequence models [37], the Transformer outperforms the Berkeley-Parser [29] even when training only on the WSJ training set of 40K sentences.

“We found this model to work poorly when we trained it on standard human-annotated parsing datasets (1M tokens)”

The source (Vinyals et al.) supports that plain RNN seq2seq models without attention do poorly when trained on only the small human-annotated WSJ data.

c36 → ref b36 · full text · Vinyals et al. 2015, Grammar as a foreign language · source (https://arxiv.org/html/1412.7449)

Supported · 7

Supported reference differs

In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

“has been predominantly used in conjunction with LSTMs”

The source says attention has predominantly been used with LSTMs, and only to a lesser extent with CNNs.

Cited reference: details differ from the found record (title).

c5 → ref b26 · full text · Parikh et al. 2016, A decomposable attention model · source (https://www.aclweb.org/anthology/D16-1244.pdf)

● **Supported** reference unchecked

We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization [1].

“we transpose batch normalization into layer normalization by computing the mean and variance used for normalization from all of the summed inputs to the neurons in a layer on a single training case”

The cited source (Ba et al., Layer Normalization) is the paper that introduces layer normalization, so it supports the use of layer normalization in the sentence.

The cited reference could not be confirmed.

c13 → ref b0 · full text · Ba et al. 2016 · source (https://arxiv.org/html/1607.06450)

● **Supported**

Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter.

“Both approaches allow us to train large LSTM networks significantly faster to the near state-of the art perplexity while using significantly less RNN parameters.”

The source supports the efficiency gains from factorization tricks (faster training, fewer parameters).

c3 → ref b20 · full text · Kuchaiev et al. 2017, Factorization tricks for LSTM networks · source (https://arxiv.org/html/1703.10722)

● **Supported**

Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter.

“these models achieve significantly better results than state-of-the-art at lower computational cost”

The source (Shazeer et al., sparsely-gated MoE) shows conditional computation yields better model performance at lower computational cost, which matches the claim for reference [32].

c3 → ref b31 · full text · Shazeer et al. 2017, Outrageously large neural networks: The sparsely-gated mixture-of-experts layer · source (https://arxiv.org/html/1701.06538)

● **Supported**

At each step the model is auto-regressive [10], consuming the previously generated symbols as additional input when generating the next.

“novel sequences can be generated from a trained network by iteratively sampling from the network’s output distribution, then feeding in the sample as input at the next step”

The source (Graves, Generating Sequences with RNNs) describes the model generating sequences by feeding each sampled output back in as input at the next step, which is the auto-regressive behavior the sentence describes.

c12 → ref b9 · full text · Graves 2013, Generating sequences with recurrent neural networks · source (https://arxiv.org/html/1308.0850)

● **Supported**

This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [38,2,9].

“The decoder decides parts of the source sentence to pay attention to.”

The source (Bahdanau et al.) introduces the encoder-decoder attention mechanism for sequence-to-sequence translation, so it is a valid example of the typical mechanism the citing sentence refers to.

c16 → ref b1 · full text · Bahdanau et al. 2014, Neural machine translation by jointly learning to align and translate · source (https://arxiv.org/html/1409.0473)

● Supported

Sentences were encoded using byte-pair encoding [3], which has a shared sourcetarget vocabulary of about 37000 tokens.

“learn shared subword units using Byte Pair Encoding (BPE) (Sennrich et al., 2016b) using 32,000 merge operations for a final vocabulary size of approximately 37k”

The source describes shared BPE subword units with a final vocabulary of about 37k, matching the sentence's claim of a shared vocabulary of about 37000 tokens.

c24 → ref b2 · full text · Britz et al. 2017, Massive exploration of neural machine translation architectures · source (https://www.aclweb.org/anthology/D17-1151.pdf)

Could not check · 5

Not enough evidence

Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38,24,15].

No verbatim quote from the source backs a verdict.

The excerpts come from Luong et al.'s attention-based NMT paper and only describe how different encoder-decoder RNN architectures differ.

c2 → ref b23 · full text · Luong et al. 2015, Effective approaches to attentionbased neural machine translation · source (https://arxiv.org/html/1508.04025)

Not enough evidence

Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4,27,28,22].

No verbatim quote from the source backs a verdict.

The excerpts come from the LSTMN paper (Cheng et al. 2016).

c8 → ref b3 · full text · Cheng et al. 2016, Long short-term memory-networks for machine reading · source (https://arxiv.org/html/1601.06733)

Not enough evidence

In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17,18] and [9].

No verbatim quote from the source backs a verdict.

The excerpts describe the Neural GPU and related architectures.

c10 → ref b16 · full text · Kaiser et al. 2016, Neural GPUs learn algorithms · source (https://arxiv.org/html/1511.08228)

Not enough evidence

Separable convolutions [6], however, decrease the complexity considerably, to O(k

No verbatim quote from the source backs a verdict.

The excerpts discuss depthwise separable convolutions and their history, but they say nothing about computational complexity, such as O(k...) or a considerable reduction.

c23 → ref b5 · full text · Chollet 2016, Xception: Deep learning with depthwise separable convolutions · source (https://arxiv.org/html/1610.02357)

Not enough evidence

In contrast to RNN sequence-to-sequence models [37], the Transformer outperforms the Berkeley-Parser [29] even when training only on the WSJ training set of 40K sentences.

No verbatim quote from the source backs a verdict.

The source is the Berkeley Parser paper (Petrov et al.), which describes the parser and its 90.2% F1.

c36 → ref b28 · abstract only · Petrov et al. 2006, Learning accurate, compact, and interpretable tree annotation · source (https://doi.org/10.3115/1220175.1220230)

References · 40

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Details differ	b26	Ankur Parikh, Oscar Täckström, Dipanjan Das, and Jakob Uszkoreit. A decomposable attention model. In Empirical Methods in Natural Language Processing, 2016.	A Decomposable Attention Model for Natural Language Inference (2016) · crossref · doi:10.18653/v1/d16-1244 title: A decomposable attention model → A Decomposable Attention Model for Natural Language Inference	Details differ from the found record (title).
Could not check	b0	Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. arXiv preprint arXiv:1607.06450, 2016.	Layer Normalization (2016) · openalex · doi:10.48550/arxiv.1607.06450	identifier resolves to 'Layer Normalization'; cited title/authors could not be extracted to compare
Verified	b1	Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. CoRR, abs/1409.0473, 2014.	Neural Machine Translation by Jointly Learning to Align and Translate (2014) · openalex · doi:10.48550/arxiv.1409.0473	identified via openalex (title + year/author); checked: title, authors (3), year
Verified	b2	Denny Britz, Anna Goldie, Minh-Thang Luong, and Quoc V. Le. Massive exploration of neural machine translation architectures. CoRR, abs/1703.03906, 2017.	Massive Exploration of Neural Machine Translation Architectures (2017) · crossref · doi:10.18653/v1/d17-1151	identified via crossref (title + year/author); checked: title, authors (5), year
Verified	b3	Jianpeng Cheng, Li Dong, and Mirella Lapata. Long short-term memory-networks for machine reading. arXiv preprint arXiv:1601.06733, 2016.	Long Short-Term Memory-Networks for Machine Reading (2016) · crossref · doi:10.18653/v1/d16-1053	identified via crossref (title + year/author); checked: arxiv_id, title, authors (3), year

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How verdicts work

References	Looked up in Crossref, OpenAlex, arXiv, Semantic Scholar and doi.org. Title, authors, year (±1) and DOI are compared by code, not by a model.
Claims	Each citing sentence is read against the cited paper: full text where open, otherwise the abstract; each row says which.
Evidence	A verdict is shown only with a quote of at least 5 words found verbatim in the source. Without one, the claim reads “Not enough evidence”, which is never counted against you.
Two readers	A judge and a skeptic on different model providers. If they disagree, the claim is flagged “Needs review” and the more cautious verdict is kept. If one of them is unavailable, the row says “judge only” or “skeptic only”.
Could not check	A lookup or model failure is never a finding: it is shown as “Could not check”, not as a problem with the manuscript.